Handbook of Clinical Neurology, Vol. 116 (3rd series)
Brain Stimulation
A.M. Lozano and M. Hallett, Editors
© 2013 Elsevier B.V. All rights reserved

Chapter 1

Principles of electrical stimulation of neural tissue

DAVID T. BROCKER AND WARREN M. GRILL*
Department of Biomedical Engineering, Pratt School of Engineering, Duke University, Durham, NC, USA

INTRODUCTION

Deep brain stimulation (DBS) is a remarkable therapy that has mainstreamed electrical stimulation of the brain for the treatment of neurological dysfunction. Before the development of DBS, neurosurgeons often resorted to ablating nervous tissue when other attempts to control the neurological dysfunction had failed. Today, DBS provides a reversible and adjustable alternative to ablative surgery. To appreciate the mechanisms of DBS, we need to understand the excitability of neural tissue. This chapter surveys the pertinent principles of electrical excitation of neural tissue, thereby providing a solid foundation from which readers can understand later chapters. While this chapter focuses on electrical stimulation in the brain, with an emphasis on DBS, many of the principles are applicable to extracellular electrical stimulation of other parts of the nervous system.

Hodgkin and Huxley's (1952) description of the ionic basis of action potentials yielded a more complete understanding of the intrinsic excitability of the neuronal membrane. However, understanding the effects of applying electrical stimulation to a population of neurons in the brain remains a challenge. The problem is complicated by the diversity of neuronal elements – soma, axons, and dendrites – and by the complexity of the brain and surrounding tissues as volume conductors. In 1975, James B. Ranck Jr published a seminal review on which neuronal elements are excited during electrical stimulation of the central nervous system (CNS). Elements of this chapter constitute a Ranck review redux, and we will re-examine and build on the topics discussed in his review in light of new research since its publication.

FUNDAMENTALS OF ELECTRICAL STIMULATION OF NEURAL TISSUE

Current passing through a resistive medium generates a distribution of electric potentials. During brain stimulation, the shape and magnitude of the potentials governs the effects of stimulation on the neural tissue. Therefore, it is important to understand three fundamental principles of electrical stimulation of the nervous system: (1) the current delivery by the stimulation electrode; (2) the electrical properties of the neural tissue medium into which the current is delivered; and (3) the impact of electrode–tissue interface. Specifically, the cumulative impedance of the DBS circuit is an important variable that can determine the total amount of current injected into the tissue, and special attention will be given to how the conductivity of the tissue affects the spread of the current in the tissue and modulates the activation of neuronal elements at a distance away from the electrode.

Current delivered to the brain tissue

A DBS system and the patient's body form a closed electrical circuit (Fig. 1.1). The implantable pulse generator (IPG) is an electrical source that causes current to flow through the extension and lead wires to the electrode, across the electrode–tissue interface, and back through the tissue to the IPG case – the return electrode. An IPG can operate as either a current source or a voltage source. During voltage-controlled DBS, the impedance of the entire circuit dictates the amount of current that flows through the neural tissue. Current-controlled stimulators deliver a constant amount of current into the neural tissue with each electrical pulse, regardless of the impedance. Typical overall impedance values range from 500 to 1500 Ω (Volkmann et al., 2002; Lempka et al., 2009; Wei and Grill, 2009). Assuming a constant impedance of 1000 Ω provides a simple rule of thumb for approximating the current delivered by voltage-controlled DBS: changing the DBS amplitude by one volt changes the DBS current amplitude by one milliamp. The resistive properties of the tissue around the electrode

*Correspondence to: Dr Warren M. Grill, Department of Biomedical Engineering, Pratt School of Engineering, Duke University, Room 136 Hudson Hall, Campus Box 90281, Durham, NC 27708-0281, USA. Tel: 919 660-5276, E-mail: warren.grill@duke.edu

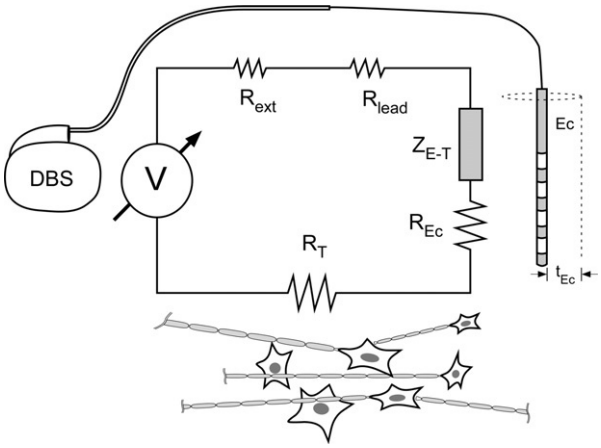

Fig. 1.1. Deep brain stimulation (DBS) circuit. The DBS system creates a closed electrical circuit with the patient's body. The amount of current delivered by voltage-controlled DBS depends on the total impedance of the circuit. There are many elements that contribute to the overall impedance: the extension wire resistance (R_{ext}); the lead wire resistance (R_{lead}); the electrode–tissue interface impedance ($Z_{E\text{-}T}$); the electrode encapsulation (E_c), thickness (t_{Ec}), and resistance (R_{Ec}); and lumped tissue resistance (R_T). The neural and nonneural tissue resistances are lumped together for simplicity. (Modified from Butson et al., 2006.)

contribute to the overall impedance (Butson et al., 2006); this tissue includes both the encapsulation tissue immediately surrounding a chronically implanted stimulation electrode, known as a glial scar, and the neural tissue.

Electrical properties of neural tissue

The resistive properties of tissue are more complicated than the lumped resistor (R_T) shown in the circuit diagram (see Fig. 1.1). Physically, resistance (R) is a measure of the voltage (V) across an element as current (I) is passed through it. The lumped resistance of that element can be defined using Ohm's law: $R = V/I$. However, for tissue with a geometric extent it is more appropriate and convenient to describe the distributed resistance of the tissue, the *resistivity*. Resistivity is denoted by ρ, with units of ohm-meter (Ω-m). In analyses of stimulation of neural tissue it is common to use *conductivity*, denoted σ, which is simply the reciprocal of resistivity with units of siemens per meter (S/m).

Conductivity, as its name suggests, quantifies the ability of a material to conduct electricity, and is useful for describing current flow in the brain (Table 1.1). Without the quasistatic approximation, however, the conductivity alone would be insufficient to describe fully the potentials generated by a current source. The quasistatic approximation allows the electrical properties of the tissue at all instants of time to be treated independently from one another. Ironically, it does not allow the brain tissue to have any "memory" of what just happened, ignoring capacitive, inductive, and wave propagation effects. Plonsey and Heppner (1967) were the first to apply this simplification to living tissues, and its use has been validated for describing potentials created by DBS with typical stimulation parameters (Bossetti et al., 2008).

Table 1.1

Important electrical properties for DBS

	Value(s)	Reference
Average brain tissue conductivity	0.15–0.3 S/m	Ranck (1963), Geddes and Baker (1967)
Cerebrospinal fluid conductivity	~1.5 S/m	Geddes and Baker (1967)
Blood conductivity*	0.6–0.7 S/m	Geddes and Baker (1967)
Encapsulation conductivity	0.05–0.2 S/m	Grill and Mortimer (1994)
Double-layer specific capacitance†	10–20 μF/cm^2	Merrill et al. (2005)
Extension and lead wire resistance	~80 Ω	Holsheimer et al. (2000b), Hemm et al. (2004)
DBS electrode impedance‡	500–1500 Ω	Volkmann et al. (2002)

*Human blood at body temperature with a normal hematocrit (40%). †Metal in aqueous solution using real area (geometric area multiplied by the roughness factor). ‡Impedances well above or below this range may indicate a mechanical failure or a short circuit in the hardware, respectively.

The brain's conductivity is inhomogeneous and anisotropic. These characteristics have been measured directly in animal models (Nicholson, 1965; Geddes and Baker, 1967) and inferred quantitatively from diffusion tensor imaging of the human brain (Tuch et al., 2001). The inhomogeneity stems from anatomical differences between regions of neural tissue (e.g., white versus gray matter) that cause differences in tissue medium conductivity (Geddes and Baker, 1967). Furthermore, neural tissue and encapsulating tissue around the chronically implanted electrode and IPG can also have different conductivities (Grill and Mortimer, 1994).

Anisotropy is the property of being directionally dependent. Neural tissue exhibits anisotropy because the axons in white matter are parallel to one another, and thus the longitudinal conductivity of white matter is greater than the transverse conductivity (Nicholson, 1965; Ranck and BeMent, 1965).

The inhomogeneity and anisotropy of brain tissue make the flow of stimulation current in the CNS

complex. Powerful finite element modeling software enables researchers to account for these properties in models of electrical stimulation of the CNS (McIntyre et al., 2004b). However, another approach is to make simplifying assumptions, for example that the tissue medium conductivity is homogeneous and isotropic. In other words, we can assign an average conductivity to the tissue medium that is not directionally dependent. These assumptions are useful for didactic purposes, as they allow the potentials generated by a current source to be calculated easily, but they ignore important contributions to the pattern of neural activation (Chaturvedi et al., 2010). The equation for the potential at a distance (r) away from a point source is given by:

$$\phi(r) = \frac{I}{4\pi\sigma r}$$

where I is the current delivered through the point source, σ is the tissue conductivity, and r is the distance from the current source. The conductivity scales how the potentials fall off as you move away from the source.

Electrode–tissue interface

The electrode–tissue interface is the site where electronic current in the electrode is transduced into ionic current in the tissue (Merrill et al., 2005). The interface has nonlinear impedance that is commonly reduced to a parallel resistor–capacitor (RC) circuit model (Fig. 1.2). The resistive element represents Faradaic charge transfer where electrons are actually passed between the electrode and the tissue during chemical reactions. The capacitive element models non-Faradaic charge transfer where no charge carriers cross the interface and currents are generated in the tissue using electromotive force to redistribute charged species.

The electrode surface area is directly proportional to the electrode's capacitance and, therefore, inversely proportional to electrode–tissue interface impedance. Hence, the electrode surface area plays a role in determining the amount of current flow through the tissue with voltage-controlled stimulation. Microelectrodes, for example, have small surface areas and little capacitance, causing them to charge/polarize quickly and inject less current into tissue. The Medtronic DBS electrode contacts have a relatively large surface area of approximately 6 mm^2 (Medtronic, 2010).

Stimulating electrodes are commonly modeled as current point sources. This is a valid approximation for sharp electrodes with sharp tips (McIntyre and Grill, 2001), but the cylindrical contacts on DBS electrodes obviously do not qualify as sharp tips. However, if interested in assessing the response of a population of neurons to stimulation, then a point source model can be used to approximate a DBS electrode contact, especially at locations removed from the conductor–insulator interfaces and the insulating electrode tip (Zhang and Grill, 2010). The point source model starts to break down close to the conductor–insulator interface because the current density distribution across the contact surface is nonuniform (Bruckenstein and Miller, 1970; Rubinstein et al., 1987; Maus et al., 1999) (see Fig 1.2). The high current density at the edges of the contact makes the perimeter of the electrode contact an important consideration for any electrode geometry.

POLARITY OF STIMULATION

The effect of stimulation does not depend exclusively on the absolute value of the stimulation current. It also depends on the current's polarity. A stimulating electrode can act as an anode (+, source of positive current) or a cathode (−, sink of positive current) at any instant in time.

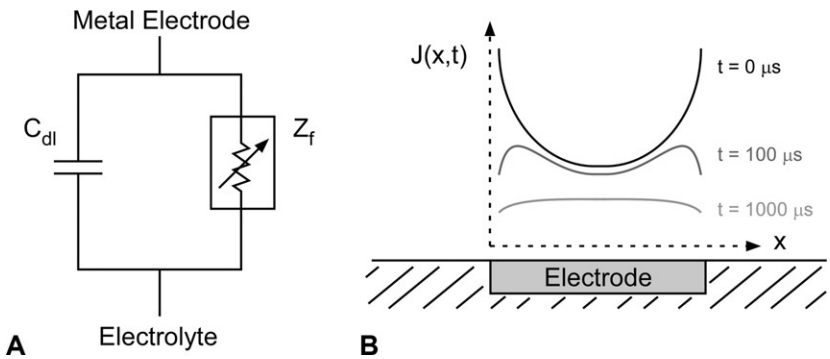

Fig. 1.2. Electrode–tissue interface. (**A**) A reduced model of the electrode–tissue interface. The electrical circuit models the paths of charge transfer at the electrode–tissue interface. Charge transfer is possible though the double-layer capacitance (C_{dl}) and Faradaic impedance (Z_f). Electrode surface area is an important consideration because the double-layer capacitance is directly proportional to the electrode surface area. (Modified from Randles, 1947.) (**B**) The current density distribution, J(x,t), across an electrode evolves during a stimulation pulse. The current density is very high at the electrode perimeter, especially early in a stimulation pulse, making the electrode perimeter an important consideration.

Extracellular cathodic stimulation primarily depolarizes neuronal membranes

As a cathode is a current sink, cathodic current is negative by convention. Recalling the equation for the potential generated by a point source, cathodic stimulation generates negative potentials in the tissue. Negative potentials in the tissue actually excite neurons more easily than positive potentials. The neuronal membrane is slightly polarized at rest. The transmembrane potential – defined as the difference between the intracellular and extracellular potential – is usually around –70 mV. When cathodic stimulation generates negative potentials in the extracellular medium, the intracellular potential is no longer negative compared with the extracellular potential; the membrane is *depolarized*. If the depolarization is strong enough or lasts long enough, the ion channel dynamics cause an action potential; the neuron is "excited" by the cathodic stimulation.

This simple explanation can be extended to include current flow as well. Because a cathode is a current sink, a cathodic stimulation pulse will draw current out of the neuron and toward the electrode. Current flow out of the neuronal membrane – that acts like a parallel RC circuit – sets up a potential across the membrane. The transmembrane potential is depolarized, and ion channel dynamics make an action potential likely.

It is important to differentiate between electronic and ionic current. We have been discussing electronic current. Electronic current generated by stimulation has to flow through the membrane and its impedance. Therefore, electronic current out of the neuronal membrane depolarizes it, and electronic current into the neuron hyperpolarizes it. On the other hand, ion channels can allow ionic current flow through the neuronal membrane with virtually no resistance. For example, during an action potential sodium cations flow into the neuron to depolarize it. The depolarization is a direct result of the transfer of positively charged ions into the cell. Because the open ion channels have essentially no resistance, there is no potential drop across the membrane arising from the flow of ionic current through the membrane.

This explanation serves to illustrate how extracellular cathodic stimulation usually excites neural tissue more easily than anodic stimulation. Both microstimulation studies in cat dorsal column and cortex (Stoney et al., 1968; BeMent and Ranck, 1969) and computational modeling (Rattay, 1998; McIntyre and Grill, 1999) suggest that cathodic stimulation is four to five times more effective, on average, at stimulating axons than anodic stimulation. This has practical implications for DBS, which presumably relies on stimulating neuronal elements to have its effects. For example, Benabid and colleagues (1996) reported that cathodic DBS suppresses tremor more effectively than anodic DBS with the same amplitude (Fig. 1.3).

The ratio between the anodic and cathodic stimulation thresholds varies widely and is strongly dependent on the position and orientation of the neuron relative to the electrode (Rattay, 1998). In fact, as Ranck points out, anodic stimulation on the surface of the cortex can be more effective than cathodic stimulation (Hern et al., 1962; Gorman, 1966; Wongsarnpigoon and Grill, 2011). Most pyramidal neurons in the cortex are oriented with their dendrites pointing toward the cortical surface. Current from the anode enters the dendrites and soma, and exits through the axon, causing depolarization of the axon.

Triphasic pattern of polarization

The difference in threshold amplitude between anodic and cathodic extracellular stimulation is a result of a triphasic pattern of polarization that is generated by extracellular stimulation. Ranck explained the triphasic pattern of polarization in terms of current flow. Figure 1.4 illustrates

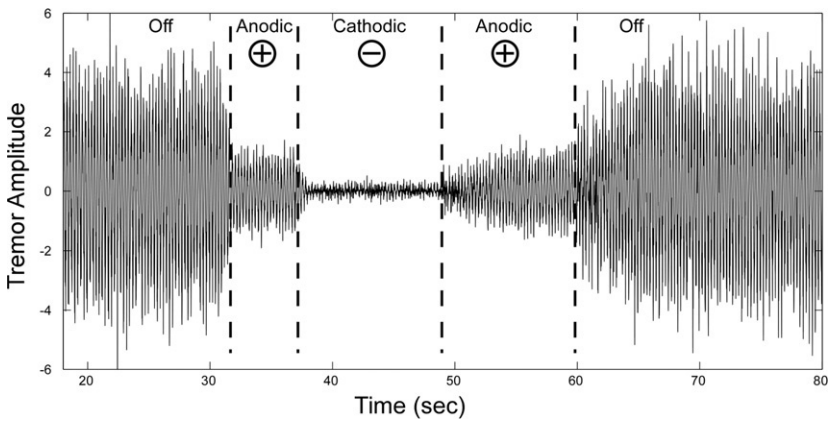

Fig. 1.3. Cathodic versus anodic stimulation for tremor suppression. High-frequency cathodic stimulation suppresses tremor more effectively than high-frequency anodic stimulation of the same magnitude. (Modified from Benabid et al., 1996.)

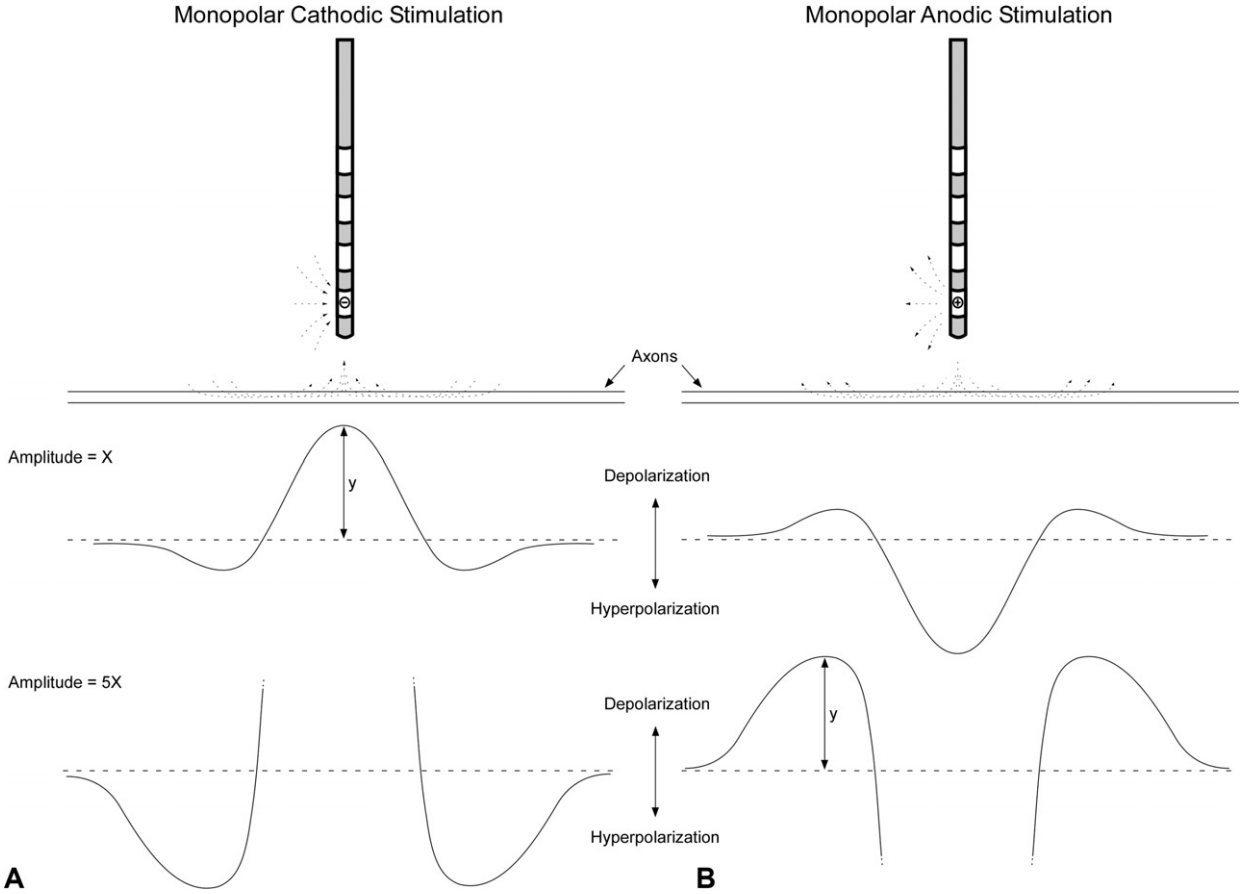

Fig. 1.4. Triphasic pattern of polarization. (**A**) A cathode (⊖) is a current sink. It draws current from the segment of the axon nearest the electrode, depolarizing it. Conservation of current dictates that current must also flow into surrounding segments of the axon, hyperpolarizing the adjacent regions. (**B**) An anode (⊕) is a current source. Anodic stimulation hyperpolarizes the region of the axon nearest to the electrode. Depolarization occurs in the adjacent regions. The depolarization magnitude is approximately five times smaller than the magnitude of the primary hyperpolarization. Therefore, if cathodic stimulation with magnitude X is able to initiate an action potential in an axon, anodic stimulation with magnitude 5X should also be able to generate an action potential by relying on the virtual cathodes.

Ranck's explanation of this phenomenon in an axon during cathodic and anodic stimulation. The section of the axon closest to a cathode experiences current flow out of the membrane, and current flows into the axon in flanking regions to satisfy conservation of current. Therefore, the axon is depolarized closest to the cathode and hyperpolarized in the flanking regions. These hyperpolarized flanking regions are often called "virtual anodes" (Basser and Roth, 2000). The currents and changes in transmembrane voltage are reversed when anodic stimulation is applied. Hyperpolarization occurs in the region of the axon closest to the anode, and the flanking regions are depolarized. Anodic stimulation relies on the flanking regions of depolarization – the "virtual cathodes" – to excite axons. Depolarization caused by anodic stimulation's virtual cathodes is usually four or five times weaker than the primary depolarization caused by cathodic stimulation of the same amplitude.

Following Ranck's explanation of the triphasic pattern of polarization using electronic current flow, investigators developed more quantitative methods for estimating the excitation of neuronal elements. The triphasic pattern of polarization can also be explained in terms of the potentials generated by stimulation using the activating function. The activating function is the second spatial derivative of the extracellular potentials ($\Delta^2 V_e$) in the direction of the axon (Rattay, 1986). The activating function for nonmyelinated axons (like the one shown in Figure 1.5) is a continuous function. However, for myelinated axons the activating function is a discrete function – the second spatial difference of the extracellular potentials – because only the potentials at the nodes of Ranvier matter. The activating function is a quite rigorous approximation of neural excitation as it accounts for the extracellular conductivity, the distance to the electrode, and the strength of stimulation,

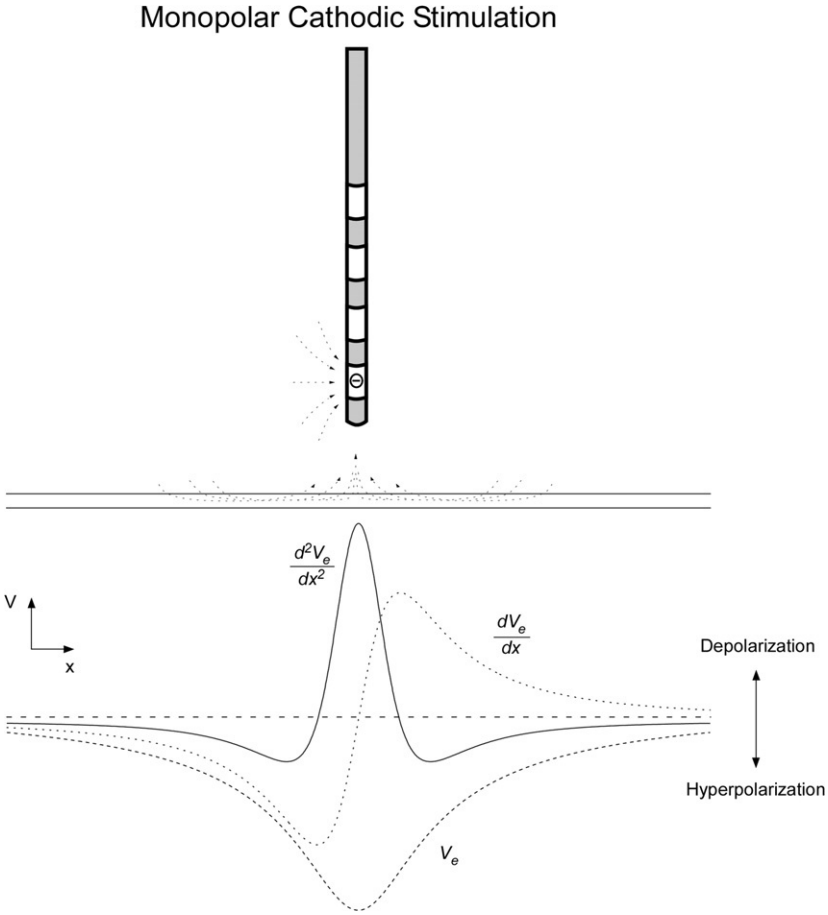

Fig. 1.5. Activating function. The activating function is a simple estimator of neural polarization. It is defined as the second spatial derivative of the extracellular potentials generated by the stimulation. The illustration shows that the activating function reproduces the triphasic pattern of polarization along an axon during cathodic stimulation.

and it is commonly used to estimate the effects of extracellular stimulation.

Stimulus waveform

Ranck did not include a discussion of stimulus waveform, but we will include it here for several reasons. First, concerns about tissue and electrode damage have motivated all chronic stimulation of the nervous system to use charge-balanced, biphasic stimulation waveforms. Second, researchers have manipulated the stimulation waveform to increase its selectivity. Lastly, waveform efficiency has also been an area of active research.

Charge-balanced stimulation

Chronic stimulation waveforms should be charge-balanced to reduce the possibility of electrode degradation and tissue damage (Lilly et al., 1955). Consider a train of high-frequency *monophasic* current pulses (Fig. 1.6). They occur so quickly in succession that the electrode does not have time to discharge completely

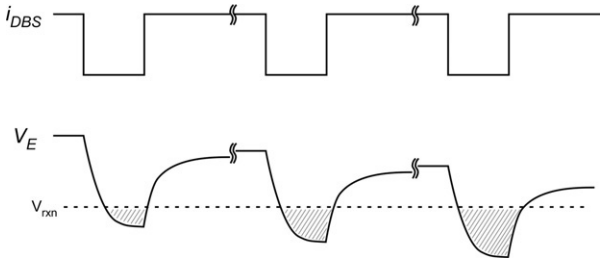

Fig. 1.6. Electrode potential drift during high-frequency stimulation. Stimulation pulses can occur so rapidly in succession during high-frequency stimulation that the electrode is unable to discharge completely between pulses. This can lead to the accumulation charge on the electrode and drift of the electrode potential. This is not desirable because it increases the driving force for irreversible chemical reactions at the electrode–tissue interface that can degrade the electrode or damage the neural tissue. Biphasic stimulation waveforms help prevent charge accumulation on the electrode. (Modified from Merrill et al., 2005.)

between pulses. The electrode potential begins to drift or "ratchet" away from the open-circuit electrode potential. The electrode potential may reach levels that provide the necessary driving force for electrochemical reactions that irreversibly degrade the electrode and generate harmful chemical species that can cause tissue damage. Even with charge-balanced biphasic waveforms, Faradaic reactions during the waveform can still cause electrode potential drift to occur. The drift is not as severe because the second, charge-balancing phase immediately moves the electrode potential back very close to the open-circuit potential. However, as Faradaic reactions may have occurred during the first phase, the second phase is not guaranteed to return the electrode to the prepulse electrode potential (Merrill et al., 2005).

Waveforms for selective stimulation

Stimulation waveform shape may allow selective stimulation of specific neuronal elements more readily than other neuronal elements. In 1975, Ranck surveyed the available evidence and suggested that, even during stimulation near a neuron's soma, the axon (including the initial segment) is stimulated. Evidence has accumulated since that time supporting that the site of action potential initiation during extracellular stimulation is the axon (Nowak and Bullier, 1998a, b; McIntyre and Grill, 1999; McIntyre et al., 2004a; Tehovnik et al., 2006). This fact can be very useful for determining the spatial effects of stimulation, as illustrated by Histed and colleagues (2009). They used cortical microstimulation and two-photon calcium imaging in the cortex of mice, rats, and cats. Instead of finding that stimulation led to a sphere of activated neurons around the microelectrode tip, they found a sparse pattern of neuron soma activation when they applied stimulation slightly above threshold. Furthermore, when they shifted the electrode slightly, different somas showed activation suggesting that the somas being activated had processes, namely axons, near the tip of the electrode. The axons closest to the electrode tip were being activated and conducting antidromic action potentials back to the soma, as well as orthodromic action potentials to their terminations.

Computational studies also support that the site of action potential initiation is always the axon or the initial segment, rather than the cell body (McIntyre and Grill, 1999; McIntyre et al., 2004a). Computational experiments were useful for fully describing this phenomenon because they allowed for precise control of parameters that are nearly impossible to control *in vivo*. The fact that action potential initiation occurs in the axon is crucial to understanding the effects of DBS (McIntyre et al., 2004a, c).

If extracellular stimulation initiates action potentials in axons, how can we expect selectively to stimulate specific neural populations? The key is to recognize the difference between axons of passage and the local projection axons. Axons of passage arise from distant cell groups and pass close by the electrode. On the other hand, local projection axons emerge from cell bodies in the vicinity of the electrode. When these axons are stimulated, the effect is the same as when the action potential initiated in the cell body.

McIntyre and Grill (2000) used this difference between axons of passage and local projection axons in a computational model of stimulation of a population of neurons to demonstrate that stimulation can be selective for specific neuronal elements, and these predictions were corroborated recently with experimental measurements (Wang et al., 2011). Monophasic anodic pulses activated a greater proportion of local projection axons at a given intensity than axons of passage. Conversely, monophasic cathodic pulses activated a greater proportion of the axons of passage. Further, charge-balanced stimulus waveforms were designed with asymmetrical phases to augment the selectively of stimulation (McIntyre and Grill, 2000). Currently, DBS devices are designed to deliver asymmetrical biphasic stimulation pulses with a leading cathodic phase followed by a longer, change-balancing, anodic phase. The selectivity of this waveform is no different from monophasic cathodic waveforms, because the low-amplitude anodic phase lags the cathodic phase and cannot exert a conditioning effect.

Although not currently implemented in DBS devices, selective stimulation of neuronal elements could be advantageous in practice. DBS and other stimulation-based neural prosthetics in the CNS are challenged by the fact that target cell bodies lie in close proximity to passing axons that arise from distant regions of the brain (Ranck, 1975; Jensen et al., 2003; Schiefer and Grill, 2006; Behrend et al., 2009; Histed et al., 2009). Consider DBS in the internal globus pallidus (GPi), where one presumably wants to stimulate the GPi neurons without stimulating fibers of the optic tract passing by the ventral border of the GPi. Selective stimulation of cell bodies in the vicinity of the electrode could enable therapeutic effects without inducing stimulation-induced visual side effects such as phosphenes.

Efficient waveforms

The shape of the stimulation waveform also affects the charge-, power-, and energy-efficiency of stimulation. Charge-efficient waveforms are desirable because the degree of tissue damage is related to the amount of charge injected (Yuen et al., 1981; McCreery et al., 1990; Shannon, 1992). Waveforms can also be power-efficient. The amount of power a stimulator needs to deliver dictates its battery size, with higher power requiring a larger battery. Lastly, energy-efficient waveforms can prolong the

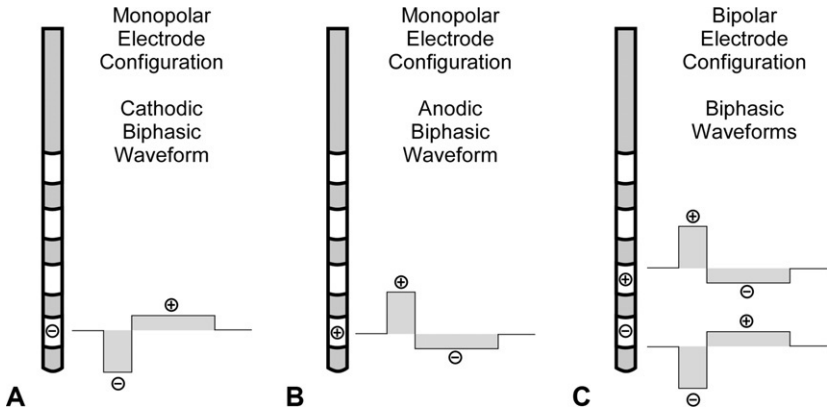

Fig. 1.7. Deep brain stimulation (DBS) electrode configurations and waveform shapes. DBS electrode contacts can be set as either an anode (⊕) or a cathode (⊖). Monopolar stimulation is delivered when only one electrode polarity is used (anodic or cathodic). Multiple active electrode contacts configured with opposite polarities deliver bipolar stimulation. (**A**) Monopolar electrode configuration with the active electrode contact configured as a cathode. The cathodic contact delivers a biphasic stimulation waveform that has a high-amplitude cathodic phase followed by a low-amplitude, charge-balancing anodic phase. (**B**) Active electrode contact configured as an anode. This configuration is not available in the clinical implantable pulse generator. (**C**) Bipolar electrode configuration. Each active electrode simultaneously delivers a biphasic waveform with the polarity of the leading phase determined by the polarity of the electrode contact.

life of implantable pulse generators because IPG lifetime is correlated linearly with energy consumption (Bin-Mahfoodh et al., 2003; Anheim et al., 2007).

Researchers have designed waveforms that are more energy-efficient than the rectangular waveforms used for DBS. Analytical studies propose that a rising exponential waveform is the most energy-efficient waveform (Offner, 1946; Jezernik and Morari, 2005). However, these analytical solutions did not take into account the nonlinear nature of the membrane dynamics, and more realistic modeling and *in vivo* experimental results demonstrated that the rising exponential waveform is no more efficient than other conventional waveform shapes (Wongsarnpigoon et al., 2010). Furthermore, coupling a realistic computational model of extracellular stimulation with engineering optimization yielded waveforms that *in vivo* experiments confirmed are more energy-efficient than other conventional waveform shapes (Wongsarnpigoon and Grill, 2010). Furthermore, Foutz and McIntyre (2010) supported this conclusion, using a detailed computational model of DBS, by concluding that Gaussian and centered-triangular waveforms are more energy-efficient than rectangular pulses.

Bipolar stimulation

Thus far we have considered only monopolar stimulation. Monopolar DBS is applied by allowing the device's metal case implanted in the patient's chest to act as the return electrode. It is essentially an infinite distance away from the neuronal elements that are stimulated, and is therefore considered monopolar stimulation. The DBS electrodes can also be configured for bipolar stimulation with two or more active electrode with opposite polarities in the brain (Fig. 1.7).

Investigators have generally assumed that bipolar stimulation is more focused than monopolar stimulation because current flow is "steered" by the presence of the electrode of the opposite polarity. The profile of polarization is definitely more complex for bipolar stimulation, but computational models suggest that the volume of tissue activated is not dramatically different (McIntyre et al., 2004b). This result is consistent with earlier work that concluded that microelectrodes in a bipolar configuration gave little reduction in current spread compared with a monopolar microelectrode (Bagshaw and Evans, 1976). One advantage of bipolar DBS is that it reduces the size of the stimulation artifact, and thereby facilitates simultaneous electrocardiography, electroencephalography, or polysomnography (Frysinger et al., 2006).

CURRENT–DISTANCE RELATIONSHIP

As extracellular potentials decline with the distance away from a source, the current needed to excite an axon depends on the distance of the axon from the stimulating source. We are interested in determining how much current is needed to excite an axon at a given distance away from an electrode. For simplicity, this discussion will be restricted to monophasic, cathodic stimuli.

Current–distance relationship definition

Stoney et al. (1968) first derived a formal current–distance relationship. The equation for threshold current as a

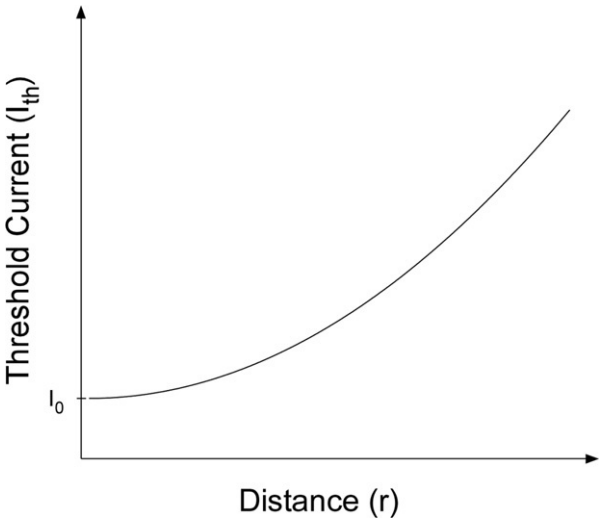

Fig. 1.8. Current–distance relationship. The current needed to excite a neuronal element increases with the square of its distance away from the stimulating electrode.

function of distance from the electrode had the following form:

$$I_{th}(r) = I_0 + Kr^2$$

where r is the distance between the electrode and the axon, and I_o and K are constants (Fig. 1.8). Even if one were able to get an electrode right next to an axon ($r = 0$), there still needs to be a small amount of current (I_o) introduced to initiate an action potential. As the electrode moves away from the axon, the threshold current increases with the square of the distance. Other researchers confirmed this current–distance relationship (Ranck, 1975; Bagshaw and Evans, 1976).

Dependence on axon diameter

The parameter K controls how quickly the threshold current increases as the electrode is moved away from the axon. K does not have the same value for all axons, but rather depends on the diameter of the axon being stimulated (Stoney et al., 1968; BeMent and Ranck, 1969; Jankowska and Roberts, 1972; Roberts and Smith, 1973). For an axon a given distance from an electrode, the greater its conduction velocity the less current required to stimulate it. Conduction velocity is directly proportional to axon diameter. Therefore, as the diameter of the axon increases, K decreases, causing the threshold current to not rise as quickly as the electrode is moved away from the axon. Generally, large-diameter axons are more easily stimulated than small-diameter axons.

Center–surround concept

The current–distance relationship describes the threshold current as a function of the axon's distance from the stimulating electrode. However, in practice we are rarely concerned with stimulating a particular axon at a given distance from the electrode with exactly the threshold current. Instead, neural prostheses, like DBS, stimulate a population of neurons around the electrode. Usually the stimulation current is well above threshold for most of the neuron excited by the stimulation, especially those in the immediate vicinity of the electrode.

Stimulation with cathodal currents much greater than threshold can actually block the propagation of action potentials in an axon. Experiments in the cat CNS showed that cathodal currents greater than about eight times threshold did not produce a propagating action potential in myelinated axons (Roberts and Smith, 1973). Similar findings were reported in cat spinal cord (Jankowska and Roberts, 1972) and frog nonmyelinated peripheral nerves (Katz and Miledi, 1965). Action potential propagation is blocked because the hyperpolarizing virtual anodes from the triphasic pattern of polarization become large enough to squelch action potential propagation (see Figs 1.4 & 1.5).

Ranck (1975) introduced the center–surround concept to describe the spatial extent of cathodal stimulation of the CNS, taking into account the fact that current many times the threshold can block action potential propagation.

STRENGTH–DURATION RELATIONSHIP

The minimum stimulus amplitude required to excite an axon a given distance from the stimulating electrode depends on the duration of the stimulation pulse. Weiss (1901) experimentally derived a linear relationship between the charge required for excitation and pulse duration. Weiss's strength–duration equation described the threshold charge as a function of pulse width. It had the form:

$$Q_{th}(PW) = I_{rh} \cdot PW + T_{ch} \cdot I_{rh}$$

where I_{rh} is the rheobase current, T_{ch} is the chronaxie, and PW is the pulse width, although Weiss did not yet recognize that the rheobase current and the chronaxie were the constants in his equation. The rheobase current is defined as the threshold current for infinitely long pulses. The chronaxie is defined as the pulse duration required for excitation when the current amplitude is equal to twice the rheobase current. Figure 1.9 illustrates the strength–duration relationship.

Lapicque (1909) reiterated Weiss's equation for the strength–duration relationship, but in terms of the threshold current, and introduced the rheobase current and chronaxie as the constants.

$$I_{th}(PW) = I_{rh}\left(1 + \frac{T_{ch}}{PW}\right)$$

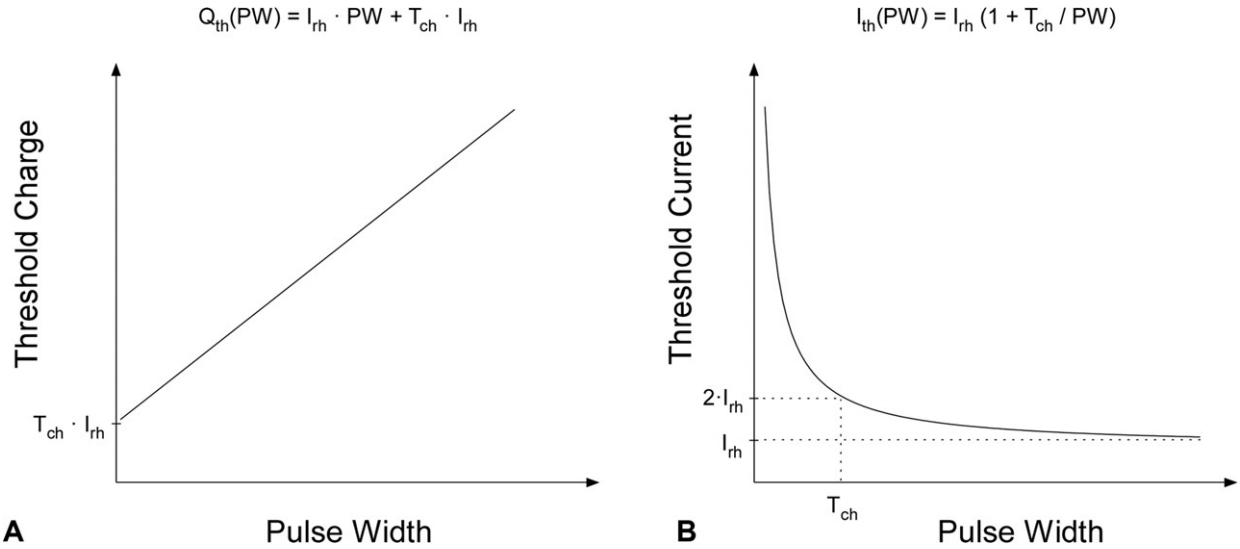

Fig. 1.9. Strength–duration relationship. Consider monophasic rectangular current pulses with variable pulse widths (PW). (**A**) The relationship between threshold charge (Q_{th}) and pulse width is linear. (**B**) The threshold current amplitude (I_{th}) decreases as the pulse duration increases in a nonlinear fashion. The minimum current that can elicit an action potential (as the pulse duration goes to infinity) is called the rheobase current (I_{rh}). The chronaxie (T_{ch}) is the shortest pulse width that can elicit an action potential when the current amplitude is twice the rheobase current.

This form of the equation is perhaps more intuitive because it describes how the threshold stimulus current amplitude decreases as the pulse width increases. As charge is the time-integral of current, for a rectangular stimulation pulse, the charge is the product of stimulation current amplitude and pulse width. Therefore, the relationship between the equations for threshold charge and threshold current is elementary; multiplying both sides of the threshold current equation by pulse width yields the threshold charge equation.

Implications for stimulus parameter selection

Clinicians are faced with a huge number of possible combinations of DBS parameters. Fortunately, the strength–duration relationship informs DBS pulse width selection by describing how the pulse width affects charge-, power-, and energy-efficiency. The equations for the power (P) and energy (E) of rectangular (monophasic) pulses, where I is the current pulse amplitude, R is the electrode impedance, and PW is the pulse width, are

$$P = I^2 \cdot R$$

$$E = I^2 \cdot R \cdot PW$$

Short pulses minimize the charge required for excitation (see Fig. 1.9). However, very long pulses minimize threshold stimulus power because instantaneous power is proportional to the square of the current amplitude. Clearly, conflict arises when trying to be simultaneously charge- and power-efficient. The energy of a rectangular current pulse is the product of its power and pulse width. Therefore, minimizing energy is analogous to finding the best compromise between charge- and power-efficiency. It turns out that stimulation is most energy-efficient when the pulse width is equal to the chronaxie. Therefore, an implantable pulse generator will last the longest when the pulse width is equal to the chronaxie of the neurons.

Implications for determining which neural elements are excited by extracellular stimulation

Chronaxie values are dependent on the type and size of the neuronal elements where the action potential is initiated. Axon chronaxies are generally in the range of 30–200 μs for large myelinated fibers and 200–700 μs for small myelinated fibers (Ranck, 1975). Therefore, rationale exists for the typical range of DBS pulse widths: 60–150 μs. The chronaxie of neuron cell bodies have been measured with intracellular micropipettes to be in the 1–10-ms range (Ranck, 1975).

Researchers have used the different chronaxies of neuronal elements to try to determine which neuronal elements they are exciting. This approach works for differentiating between large- and small-diameter axons, but it cannot be applied to differentiate between excitation of cell bodies and axons because action potential initiation always occurs in axons. More specifically, the chronaxie from a strength–duration curve measured with extracellular stimulation cannot tell you whether

local cell bodies or axons of passage are excited in the vicinity of the stimulating electrode (Miocinovic and Grill, 2004). Chronaxie measurements imply that the excitation of large-diameter axons is responsible for the effects of DBS in the subthalamic nucleus (STN) (Ashby et al., 1999, 2001), GPi (Holsheimer et al., 2000a; Wu et al., 2001), and thalamus (Strafella et al., 1997; Holsheimer et al., 2000a), but these could indeed be the axons of local neurons (McIntyre et al., 2004a).

FREQUENCY OF STIMULATION

Ranck did not discuss how neurons respond to changes in the frequency of electrical stimulation, but these considerations are important when characterizing neural excitability. The border between "high frequency" and "low frequency" DBS is usually considered to be 100 Hz. Suprathreshold cathodic current pulses applied at a low frequency initiate action potentials in neurons with high fidelity. Neurons respond to the stimuli similarly to how they would respond to a single stimulus pulse because of the relatively long interpulse interval (IPI).

High-frequency stimulation

High-frequency stimulation can also initiate action potentials in axons with high fidelity. As the IPI is short, action potentials generated by high-frequency stimulation are far more prevalent than intrinsic ones. Therefore, high-frequency stimulation can mask intrinsic activity (Grill et al., 2004), which may be desirable. The refractory period limits how rapidly neurons can fire action potentials in response to high-frequency stimulation, no matter how strong the stimuli are. The refractory period ($\sim$1 ms) is the time immediately following an action potential when ion channel dynamics make it impossible for the neuron to fire another action potential; if the IPI is shorter than the refractory period, the neuron will not be able to respond to each stimulus with an action potential.

High-frequency ($>$100 Hz) DBS is effective for movement disorders. In patients with Parkinson's disease, STN-DBS frequencies above 100 Hz lead to the greatest improvement in tremor, akinesia, and rigidity (Rizzone et al., 2001; Moro et al., 2002). There is some evidence to suggest that there exists an upper limit to the therapeutic frequency range, possibly around 250 Hz (Rizzone et al., 2001; Moro et al., 2002). Frequencies below 50 Hz do not ameliorate parkinsonian signs (Rizzone et al., 2001; Moro et al., 2002). Furthermore, low-frequency stimulation can actually worsen some symptoms of Parkinson's disease (Moro et al., 2002). High-frequency DBS is also necessary for suppression of tremor in patients with essential tremor (Benabid et al., 1996), and low-frequency stimulation can exacerbate essential tremor (Kuncel et al., 2006). Understanding why DBS is therapeutic when applied within a specific range of frequencies could help explain its mechanisms of action.

Temporally nonregular stimulation, with IPIs that vary from one pair of pulses to the next, can be used to explore the mechanisms of DBS (Birdno et al., 2008, 2012; Dorval et al., 2010). Temporally, nonregular DBS patterns created from Gaussian distributions of IPIs ameliorated motor symptoms less effectively than temporally regular stimulation in patients with Parkinson's disease and essential tremor (Birdno et al., 2008; Dorval et al., 2010), highlighting the importance of regularizing neural activity (Grill et al., 2004).

Stimulation frequency and synaptic transmission

Nervous system function relies on a network of neurons communicating with one another via neurotransmitter release at synapses. Electrical stimulation in the CNS can change the strength of synapses at the end of the stimulated axons. Furthermore, synapses near a stimulating electrode can modulate the stimulation's effect on local neurons.

When the efficacy of synaptic transmission changes, it is called synaptic plasticity. Synaptic plasticity comes in many different forms. Long-term potentiation (LTP) and long-term depression (LTD) are types of synaptic plasticity where the change in synaptic transmission persists for a long time (hours to days) after the conditioning stimuli. The frequency of the conditioning stimuli often dictates what type of plasticity occurs. The stimulation location also plays a big role, and a specific frequency of stimulation can modulate synapses very differently depending on the region of the brain.

There are also forms of synaptic plasticity that last for only a few minutes. This short-term plasticity is also sensitive to the frequency of activity at the synapse. For example, a pair of high-frequency stimuli will tend to facilitate a synapse, causing the synapse to be stronger for the second pulse. At some synapses, a prolonged period of high-frequency activity will lead to a decrease in synaptic strength. The most widespread mechanism appears to be a decrease in the number of synaptic vesicles carrying the neurotransmitter that are available for release (Zucker and Regehr, 2002).

Further, synapses can modulate the effects of electrical stimulation in a frequency-dependent manner. This "indirect effect" is imposed on neurons around a stimulating electrode through activation of presynaptic terminals. The "direct effect" – direct excitation of the neuron's axon – still persists.

Extracellular stimulation very effectively excites axon terminals in the vicinity of the stimulating electrode

(Baldissera et al., 1972; Jankowska et al., 1975; Gustafsson and Jankowska, 1976; McIntyre and Grill, 2002). Therefore, the indirect effect of stimulation is of interest. However, it is difficult to investigate the indirect effect experimentally. Computational modeling suggests that terminal excitation does little to modulate axonal firing because action potentials are initiated in the axon during DBS (McIntyre and Grill, 2002). However, terminal excitation can help explain axon–soma decoupling. High-frequency stimulation of a region of the brain with inhibitory afferent projections causes release of inhibitory neurotransmitters on a local neuron's dendrites and soma. The soma and dendrites are hyperpolarized, but the stimulation is still initiating action potentials in the neuron's axon (McIntyre et al., 2004a). The soma activity – which is usually what is recorded from extracellular microelectrodes – is completely decoupled from the axon's activity. For this reason, interpreting microelectrode recording near extracellular stimulation can be difficult.

CONSIDERATIONS FOR NONDAMAGING STIMULATION

Introducing a stimulating electrode into the brain will cause mechanical damage to a degree related to the electrode size. The electrode should be made of nontoxic materials that are resistant to electrochemical dissolution. Even so, the presence of an electrode will incite a foreign body response that can damage or kill neurons around the electrode (Biran et al., 2005). Mechanical damage cannot be avoided when inserting an electrode into the brain, but electrode modifications could possibly mitigate the foreign body response.

Damage to the tissue can also occur by delivering electrical stimulation through the electrode. Therefore, clinicians need to be aware of how stimulation parameters can affect tissue damage. There are two proposed mechanisms for stimulation-induced tissue damage: the evolution of toxic products at the electrode–tissue interface and physiological mass action.

Toxic electrochemical reactions

The electrode–tissue interface is where electronic charge carriers in the electrode are transduced into ionic charge carriers in the tissue. The resistive element of the interface impedance (see Fig. 1.2) represents Faradaic charge transfer where electrons are actually passed between the electrode and the tissue during chemical reactions. These chemical reactions can degrade the electrode and cause the evolution of toxic products that diffuse into the surrounding tissue. Tissue damage will ensue if the rate of toxic product generation at the interface is greater than that which the tissue can tolerate.

Differences in the amount of damage caused by monophasic and charge-balanced biphasic stimulation waveforms support that the creation of toxic products at the interface is a mechanism of stimulation-induced tissue damage. Multiple experiments have demonstrated that a monophasic stimulation protocol no longer causes tissue damage if a charge-balanced waveform is used (Lilly et al., 1955; Mortimer et al., 1970; Pudenz et al., 1975), suggesting that the evolution of toxic products at the interface is damaging the neurons during monophasic stimulation. The undesirable reactions that can occur during stimulation include electrolysis of water; oxidation of saline, metal, and organic materials; and reduction of oxygen (Brummer and Turner, 1977; Morton et al., 1994).

Physiological mass action

Tissue damage may also be caused by physiological changes associated with stimulation-induced neural excitation. This mechanism of tissue damage is linked with synchronous activity in large populations of neurons, and can be thought of as the result of overstimulation of excitable tissue. McCreery and colleagues (1988) presented a convincing argument that mass action is a mechanism of stimulation-induced injury. They stimulated the cat cortex with identical stimulation protocols but with two types of electrode: platinum and sintered tantalum pentoxide electrodes. Platinum electrodes can participate in Faradaic charge transfer reactions, whereas sintered tantalum pentoxide electrodes engage only in capacitive charge transfer. Tissue damage occurred around both types of electrode, suggesting that the damage originated from the actual stimulation of the tissue, rather than solely from the creation of toxic products from electrochemical reactions.

How mass action leads to neuronal injury is still unclear. An experiment that blocked neuronal activity with local anesthetic in a stimulated peripheral nerve prevented the mass action injury that otherwise occurred in the unblocked nerve (Agnew et al., 1990). This suggests that action potential firing is necessary for mass action injury. There is also evidence from CNS stimulation that mass action is similar to excitotoxicity. In excitotoxicity, excessive release of the excitatory neurotransmitter glutamate causes a pathological increase in intracellular calcium concentrations, which leads to cell death. Calcium can enter a neuron through glutamate receptors, *N*-methyl-D-aspartate (NMDA) receptors, and an NMDA receptor antagonist can protect against neural injury induced by electrical stimulation in the cat cortex (Agnew

et al., 1993). Strong electrical stimulation in the CNS could cause massive glutamate release and excitotoxicity. Other proposed mechanisms of mass action of induced neural injury include changes in intracellular and extracellular ionic concentrations and a depletion of oxygen or glucose (Merrill et al., 2005).

The mass action and toxic electrochemical reactions mechanisms are probably not mutually exclusive (McCreery et al., 1992; Agnew et al., 1993; Merrill et al., 2005). The neuronal injury around any stimulating electrode is probably a result of a combination of mechanical damage, the foreign body response, the creation of toxic electrochemical reaction products, and physiological mass action.

Nondamaging stimulation intensities

The degree of stimulation-induced neuronal injury is determined by two cofactors: the charge density and the charge per phase (McCreery et al., 1990). Charge is the time-integral of current and, for a monophasic, rectangular stimulation pulse, the charge per phase is the product of stimulation current amplitude and pulse width. The average charge density is the charge per phase divided by the electrode surface area. Shannon (1992) proposed a neural damage model describing the relationship between these factors:

$$\log(D) = k - \log(Q)$$

where D is the charge density per phase, Q is the charge per phase, and k ($\sim$1.85) is an experimentally derived constant that defines the boundary between stimulation that produced tissue damage and stimulation that did not (Fig. 1.10). It is important to note that Shannon's model may not be directly applicable to DBS because the data used by Shannon to delineate between damaging and nondamage stimulation intensities did not come from studies using DBS electrodes, DBS target brain areas, or typical DBS frequencies.

The two factors for determining the threshold for neural injury by stimulation match well with the proposed mechanisms of neural injury. The charge per phase is related to physiological mass action because it determines the amount and spatial extent of stimulation-induced excitation (see Fig. 1.9). The charge density per phase is related to damage from toxic electrochemical products, because it influences the type and rate of electrochemical reactions that occur at the electrode surface.

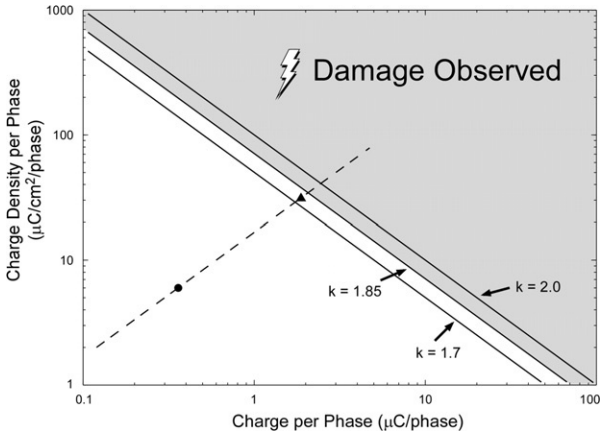

Fig. 1.10. Shannon model of neuronal damage. Assuming a constant impedance of 1 kΩ, the dotted line shows the full range of Medtronic deep brain stimulation (DBS) parameters. The triangle (▲) represents Medtronic's recommended charge density limit (30 μC/cm^2) based on the Shannon model. Medtronic DBS systems prohibit some combinations of parameters with charge densities above 30 μC/cm^2 on the dotted line (Medtronic, 2006). The circle (●) represents a typical DBS parameter setting (90 μs, 3 volts). Post-mortem DBS-induced tissue damage has not been observed with DBS parameters near this typical DBS parameter setting (Haberler et al., 2000; Burbaud et al., 2002; Henderson et al., 2002). (Modified from Shannon, 1992.)

CLOSING REMARKS

Electrical stimulation of the brain is a complex and multifaceted problem that remains a challenge to describe fully. The diversity of neuronal elements and the complexity of neural tissue as a volume conductor do not allow simple generalizations of the effects of stimulation on populations of neurons in the brain. Instead, a strong foundation in the principles described in this chapter facilitates understanding the complex effects of electrical stimulation in the brain.

References

Agnew W, McCreery D, Yuen T et al. (1990). Local anaesthetic block protects against electrically-induced damage in peripheral nerve. J Biomed Eng 12: 301–308.

Agnew WF, McCreery DB, Yuen TGH et al. (1993). Mk-801 protects against neuronal injury induced by electrical stimulation. Neuroscience 52: 45–53.

Anheim M, Fraix V, Chabardès S et al. (2007). Lifetime of Itrel II pulse generators for subthalamic nucleus stimulation in Parkinson's disease. Mov Disord 22: 2436–2439.

Ashby P, Kim YJ, Kumar R et al. (1999). Neurophysiological effects of stimulation through electrodes in the human subthalamic nucleus. Brain 122: 1919–1931.

Ashby P, Paradiso G, Saint-Cyr J et al. (2001). Potentials recorded at the scalp by stimulation near the human subthalamic nucleus. Clin Neurophysiol 112: 431–437.

Bagshaw E, Evans M (1976). Measurement of current spread from microelectrodes when stimulating within the nervous system. Exp Brain Res 25: 391–400.

Baldissera F, Lundberg A, Udo M (1972). Stimulation of pre- and postsynaptic elements in the red nucleus. Exp Brain Res 15: 151–167.

Basser PJ, Roth BJ (2000). New currents in electrical stimulation of excitable tissues. Annu Rev Biomed Eng 2: 377–397.

Behrend CE, Cassim SM, Pallone MJ et al. (2009). Toward feedback controlled deep brain stimulation: dynamics of glutamate release in the subthalamic nucleus in rats. J Neurosci Methods 180: 278–289.

BeMent SL, Ranck JB (1969). A quantitative study of electrical stimulation of central myelinated fibers. Exp Neurol 24: 147–170.

Benabid AL, Pollak P, Gao D et al. (1996). Chronic electrical stimulation of the ventralis intermedius nucleus of the thalamus as a treatment of movement disorders. J Neurosurg 84: 203–214.

Bin-Mahfoodh M, Hamani C, Sime E et al. (2003). Longevity of batteries in internal pulse generators used for deep brain stimulation. Stereotact Funct Neurosurg 80: 56–60.

Biran R, Martin DC, Tresco PA (2005). Neuronal cell loss accompanies the brain tissue response to chronically implanted silicon microelectrode arrays. Exp Neurol 195: 115–126.

Birdno M, Kuncel A, Dorval A et al. (2008). Tremor varies as a function of the temporal regularity of deep brain stimulation. Neuroreport 19: 599–602.

Birdno MJ, Kuncel AM, Dorval AD et al. (2012). Stimulus features underlying reduced tremor suppression with temporally patterned deep brain stimulation. J Neurophysiol 107: 364–383.

Bossetti CA, Birdno MJ, Grill WM (2008). Analysis of the quasi-static approximation for calculating potentials generated by neural stimulation. J Neural Eng 5: 44–53.

Bruckenstein S, Miller B (1970). An experimental study of nonuniform current distribution at rotating disk electrodes. J Electrochem Soc 117: 1044–1048.

Brummer SB, Turner MJ (1977). Electrochemical considerations for safe electrical stimulation of the nervous system with platinum electrodes. IEEE Trans Biomed Eng 24: 59–63.

Burbaud P, Vital A, Rougier A et al. (2002). Minimal tissue damage after stimulation of the motor thalamus in a case of chorea-acanthocytosis. Neurology 59: 1982–1984.

Butson CR, Maks CB, McIntyre CC (2006). Sources and effects of electrode impedance during deep brain stimulation. Clin Neurophysiol 117: 447–454.

Chaturvedi A, Butson CR, Lempka SF et al. (2010). Patient-specific models of deep brain stimulation: influence of field model complexity on neural activation predictions. Brain Stimul 3: 65–77.

Dorval AD, Kuncel AM, Birdno MJ et al. (2010). Deep brain stimulation alleviates parkinsonian bradykinesia by regularizing pallidal activity. J Neurophysiol 104: 911–921.

Foutz TJ, McIntyre CC (2010). Evaluation of novel stimulus waveforms for deep brain stimulation. J Neural Eng 7: 066008.

Frysinger RC, Quigg M, Elias WJ (2006). Bipolar deep brain stimulation permits routine EKG, EEG, and polysomnography. Neurology 66: 268–270.

Geddes L, Baker L (1967). The specific resistance of biological material – a compendium of data for the biomedical engineer and physiologist. Med Biol Eng Comput 5: 271–293.

Gorman A (1966). Differential patterns of activation of the pyramidal system elicited by surface anodal and cathodal cortical stimulation. J Neurophysiol 29: 547–564.

Grill WM, Mortimer TJ (1994). Electrical properties of implant encapsulation tissue. Ann Biomed Eng 22: 23–33.

Grill W, Snyder A, Miocinovic S (2004). Deep brain stimulation creates an informational lesion of the stimulated nucleus. Neuroreport 15: 1137–1140.

Gustafsson B, Jankowska E (1976). Direct and indirect activation of nerve cells by electrical pulses applied extracellularly. J Physiol 258: 33–61.

Haberler C, Alesch F, Mazal PR et al. (2000). No tissue damage by chronic deep brain stimulation in Parkinson's disease. Ann Neurol 48: 372–376.

Hemm S, Vayssiere N, Mennessier G et al. (2004). Evolution of brain impedance in dystonic patients treated by GPi electrical stimulation. Neuromodulation 7: 67–75.

Henderson JM, Pell M, O'Sullivan DJ et al. (2002). Postmortem analysis of bilateral subthalamic electrode implants in Parkinson's disease. Mov Disord 17: 133–137.

Hern J, Landgren S, Phillips C et al. (1962). Selective excitation of corticofugal neurones by surface-anodal stimulation of the baboon's motor cortex. J Physiol 161: 73–90.

Histed MH, Bonin V, Reid RC (2009). Direct activation of sparse, distributed populations of cortical neurons by electrical microstimulation. Neuron 63: 508–522.

Hodgkin AL, Huxley AF (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. J Physiol 117: 500–544.

Holsheimer J, Demeulemeester H, Nuttin B et al. (2000a). Identification of the target neuronal elements in electrical deep brain stimulation. Eur J Neurosci 12: 4573–4577.

Holsheimer J, Dijkstra EA, Demeulemeester H et al. (2000b). Chronaxie calculated from current-duration and voltage-duration data. J Neurosci Methods 97: 45–50.

Jankowska E, Roberts W (1972). An electrophysiological demonstration of the axonal projections of single spinal interneurones in the cat. J Physiol 222: 597–622.

Jankowska E, Padel Y, Tanaka R (1975). The mode of activation of pyramidal tract cells by intracortical stimuli. J Physiol 249: 617–636.

Jensen RJ, Rizzo JF, Ziv OR et al. (2003). Thresholds for activation of rabbit retinal ganglion cells with an ultrafine, extracellular microelectrode. Invest Ophthalmol Vis Sci 44: 3533–3543.

Jezernik S, Morari M (2005). Energy-optimal electrical excitation of nerve fibers. IEEE Trans Biomed Eng 52: 740–743.

Katz B, Miledi R (1965). Propagation of electric activity in motor nerve terminals. Proc R Soc Lond B Biol Sci 161: 453–482.

Kuncel AM, Cooper SE, Wolgamuth BR et al. (2006). Clinical response to varying the stimulus parameters in deep brain stimulation for essential tremor. Mov Disord 21: 1920–1928.

Lapicque L (1909). Definition experimentale de l'excitabilité. C R Acad Sci 67: 280–283.

Lempka SF, Miocinovic S, Johnson MD et al. (2009). In vivo impedance spectroscopy of deep brain stimulation electrodes. J Neural Eng 6: 046001.

Lilly JC, Hughes JR, Alvord EC et al. (1955). Brief, noninjurious electric waveform for stimulation of the brain. Science 121: 468–469.

Maus RG, McDonald EM, Wightman RM (1999). Imaging of nonuniform current density at microelectrodes by electrogenerated chemiluminescence. Anal Chem 71: 4944–4950.

McCreery D, Agnew W, Yuen TGH et al. (1988). Comparison of neural damage induced by electrical stimulation with Faradaic and capacitor electrodes. Ann Biomed Eng 16: 463–481.

McCreery DB, Agnew WF, Yuen TGH et al. (1990). Charge density and charge per phase as cofactors in neural injury induced by electrical stimulation. IEEE Trans Biomed Eng 37: 996–1001.

McCreery D, Agnew W, Yuen TGH et al. (1992). Damage in peripheral nerve from continuous electrical stimulation: comparison of two stimulus waveforms. Med Biol Eng Comput 30: 109–114.

McIntyre CC, Grill WM (1999). Excitation of central nervous system neurons by nonuniform electric fields. Biophys J 76: 878–888.

McIntyre CC, Grill WM (2000). Selective microstimulation of central nervous system neurons. Ann Biomed Eng 28: 219–233.

McIntyre CC, Grill WM (2001). Finite element analysis of the current-density and electric field generated by metal microelectrodes. Ann Biomed Eng 29: 227–235.

McIntyre CC, Grill WM (2002). Extracellular stimulation of central neurons: influence of stimulus waveform and frequency on neuronal output. J Neurophysiol 88: 1592–1604.

McIntyre CC, Grill WM, Sherman DL et al. (2004a). Cellular effects of deep brain stimulation: model-based analysis of activation and inhibition. J Neurophysiol 91: 1457–1469.

McIntyre CC, Mori S, Sherman DL et al. (2004b). Electric field and stimulating influence generated by deep brain stimulation of the subthalamic nucleus. Clin Neurophysiol 115: 589–595.

McIntyre CC, Savasta M, Kerkerian-Le Goff L et al. (2004c). Uncovering the mechanism(s) of action of deep brain stimulation: activation, inhibition, or both. Clin Neurophysiol 115: 1239–1248.

Medtronic (2006). Medtronic Kinetra® dual program neurostimulator for deep brain stimulation, 7428. Technical Manual. Medtronic, Minneapolis, MN.

Medtronic (2010). Medtronic DBS™ lead kit for deep brain stimulation, 3387, 3389. Implant Manual. Medtronic, Minneapolis, MN.

Merrill DR, Bikson M, Jefferys JGR (2005). Electrical stimulation of excitable tissue: design of efficacious and safe protocols. J Neurosci Methods 141: 171–198.

Miocinovic S, Grill WM (2004). Sensitivity of temporal excitation properties to the neuronal element activated by extracellular stimulation. J Neurosci Methods 132: 91–99.

Moro E, Esselink RJA, Xie J et al. (2002). The impact on Parkinson's disease of electrical parameter settings in STN stimulation. Neurology 59: 706–713.

Mortimer JT, Shealy CN, Wheeler C (1970). Experimental nondestructive electrical stimulation of the brain and spinal cord. J Neurosurg 32: 553.

Morton SL, Daroux ML, Mortimer JT (1994). The role of oxygen reduction in electrical stimulation of neural tissue. J Electrochem Soc 141: 122–130.

Nicholson PW (1965). Specific impedance of cerebral white matter. Exp Neurol 13: 386–401.

Nowak LG, Bullier J (1998a). Axons, but not cell bodies, are activated by electrical stimulation in cortical gray matter I. Evidence from chronaxie measurements. Exp Brain Res 118: 477–488.

Nowak LG, Bullier J (1998b). Axons, but not cell bodies, are activated by electrical stimulation in cortical gray matter II. Evidence from selective inactivation of cell bodies and axon initial segments. Exp Brain Res 118: 489–500.

Offner F (1946). Stimulation with minimum power. J Neurophysiol 9: 387–390.

Plonsey R, Heppner D (1967). Considerations of quasi-stationarity in electrophysiological systems. Bull Math Biol 29: 657–664.

Pudenz R, Bullara L, Jacques S et al. (1975). Electrical stimulation of the brain. III. The neural damage model. Surg Neurol 4: 389–400.

Ranck JB (1963). Specific impedance of rabbit cerebral cortex. Exp Neurol 7: 144–152.

Ranck JB (1975). Which elements are excited in electrical stimulation of mammalian central nervous system: a review. Brain Res 98: 417–440.

Ranck JB, BeMent SL (1965). The specific impedance of the dorsal columns of cat: an anisotropic medium. Exp Neurol 11: 451–463.

Randles J (1947). Kinetics of rapid electrode reactions. Discuss Faraday Soc 1: 11–19.

Rattay F (1986). Analysis of models for external stimulation of axons. IEEE Trans Biomed Eng 33: 974–977.

Rattay F (1998). Analysis of the electrical excitation of CNS neurons. IEEE Trans Biomed Eng 45: 766–772.

Rizzone M, Lanotte M, Bergamasco B et al. (2001). Deep brain stimulation of the subthalamic nucleus in Parkinson's disease: effects of variation in stimulation parameters. J Neurol Neurosurg Psychiatry 71: 215–219.

Roberts W, Smith D (1973). Analysis of threshold currents during microstimulation of fibres in the spinal cord. Acta Physiol Scand 89: 384–394.

Rubinstein JT, Spelman FA, Soma M et al. (1987). Current density profiles of surface mounted and recessed electrodes for neural prostheses. IEEE Trans Biomed Eng 34: 864–875.

Schiefer MA, Grill WM (2006). Sites of neuronal excitation by epiretinal electrical stimulation. IEEE Trans Neural Syst Rehabil Eng 14: 5–13.

Shannon RV (1992). A model of safe levels for electrical stimulation. IEEE Trans Biomed Eng 39: 424–426.

Stoney SD, Thompson WD, Asanuma H (1968). Excitation of pyramidal tract cells by intracortical microstimulation: effective extent of stimulating current. J Neurophysiol 31: 659–669.

Strafella A, Ashby P, Munz M et al. (1997). Inhibition of voluntary activity by thalamic stimulation in humans: relevance for the control of tremor. Mov Disord 12: 727–737.

Tehovnik EJ, Tolias AS, Sultan F et al. (2006). Direct and indirect activation of cortical neurons by electrical microstimulation. J Neurophysiol 96: 512–521.

Tuch DS, Wedeen VJ, Dale AM et al. (2001). Conductivity tensor mapping of the human brain using diffusion tensor MRI. Proc Natl Acad Sci U S A 98: 11697–11701.

Volkmann J, Herzog J, Kopper F et al. (2002). Introduction to the programming of deep brain stimulators. Mov Disord 17: S181–S187.

Wang Q, Millard DC, Zheng HJV et al. (2011). Enhanced cortical specificity of thalamic microstimulation through manipulation of charge delivery. Society for Neuroscience, Washington, DC.

Wei XF, Grill WM (2009). Impedance characteristics of deep brain stimulation electrodes in vitro and in vivo. J Neural Eng 6: 046008.

Weiss G (1901). Sur la possibilité de rendre comparables entre eux les appareils servant à l'excitation électrique. Arch Ital Biol 35: 413–446.

Wongsarnpigoon A, Grill WM (2010). Energy-efficient waveform shapes for neural stimulation revealed with a genetic algorithm. J Neural Eng 7: 046009.

Wongsarnpigoon A, Grill WM (2011). Effects of stimulation parameters and electrode location on thresholds for epidural stimulation of cat motor cortex. J Neural Eng 8: 066016.

Wongsarnpigoon A, Woock JP, Grill WM (2010). Efficiency analysis of waveform shape for electrical excitation of nerve fibers. IEEE Trans Neural Syst Rehabil Eng 18: 319–328.

Wu Y, Levy R, Ashby P et al. (2001). Does stimulation of the GPi control dyskinesia by activating inhibitory axons? Mov Disord 16: 208–216.

Yuen TGH, Agnew WF, Bullara LA et al. (1981). Histological evaluation of neural damage from electrical stimulation: considerations for the selection of parameters for clinical application. Neurosurgery 9: 292–299.

Zhang TC, Grill WM (2010). Modeling deep brain stimulation: point source approximation versus realistic representation of the electrode. J Neural Eng 7: 066009.

Zucker RS, Regehr WG (2002). Short-term synaptic plasticity. Annu Rev Physiol 64: 355–405.